PH-Corpus v0.1: a 3.19-billion-token Philippine-language corpus for continued pretraining, and how it was built, checked and registered

PinoAI data paper · CORPUS-001 · PH-Corpus v0.1 r2 (DEC-005 Amendment 12) · Taglish edition: CORPUS-001-ph-corpus-v0.1.tl.qmd

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Abstract

We describe PH-Corpus v0.1, the training corpus for PinoAI's first continued-pretraining (CPT) experiment, DEC-005. The build used here, r2, holds 3,296,100 training documents and 3,190,645,093 training tokens, counted with the Gemma 4 E4B tokenizer:

- Tagalog/Filipino web text, 75.93%;
- English replay, 15.86%;
- seven regional languages (Cebuano, Ilokano, Hiligaynon, Waray, Bikol, Kapampangan, Pangasinan), 7.27%;
- scripture in those languages, 0.86%;
- public-domain Tagalog literature, 0.09%.

It also holds two evaluation splits: 17,265 held-out documents and 13 out-of-distribution books.

Every source is licensed for commercial use. The pipeline does the following:

- redacts contact, identity and payment data;
- blocks *.gov.ph pages;
- filters low-quality and duplicate text;
- routes scripture to its own bucket, so its weight can be set separately;
- restores an archaic Tagalog spelling (g with tilde) that the plain-text source had flattened;
- removes any document that overlaps one of 57,315 benchmark items.

Two independent checks found nothing left behind. A re-scan of all 3,313,378 documents found 0 benchmark hits, and the private upload verified all 69 files by sha256. The corpus is registered by its manifest hash as DEC-005 Amendment 12.

We also report what the corpus is not:

- it is not a public release;
- its per-language labels for the small regional languages are noisy;
- it carries an unresolved privacy review, pending the data protection officer.

1. Why this corpus exists

DEC-005 asks whether continued pretraining on Philippine-language text improves the open Gemma 4 E4B base on Filipino without breaking its English. The experiment is pre-registered: every config, script and corpus is pinned by sha256 in `docs/DECISION_LOG.md` before it is used. PH-Corpus v0.1 is the data side of that test. Its design goals were:

1. **Commercial-clean.** Train only on data licensed for commercial use; non-commercial data is for evaluation only.
2. **Measured, not estimated.** Every share in the training mixture is set from the build's own token counts.
3. **No leakage into our benchmarks.** Any document that overlaps an evaluation item is removed before training.
4. **Reproducible.** One manifest records every source file's hash, every pipeline version and every counter.

2. Sources and licences

Table 1: Sources in the r2 manifest (56 files). Hugging Face sources are fetched at a pinned 40-character revision and each file is verified against its LFS sha256; a mismatch stops the build.

source	what it contributes	licence (from the source registry)	files
FineWeb-2 `fil_Latn`	Tagalog/Filipino web text	ODC-By 1.0 (packaging) + CommonCrawl Terms of Use	1
FineWeb-2 regional	ceb, hil, ilo, bcl, pam, pag, war web text	ODC-By 1.0 + CommonCrawl Terms of Use	7
FineWeb (English), sample-10BT slice	English replay	ODC-By 1.0 + CommonCrawl Terms of Use	15
Project Gutenberg, Tagalog allow-list	public-domain literature	public domain in the Philippines for the allow-listed works (RA 8293 s.213.1, s.214)	32
Ang Biblia (1905)	public-domain Tagalog Bible	Public Domain (CrossWire module licence)	1

The web-crawl licences cover the *collection*. The page text still belongs to each website's owner. Accepting that risk, which is standard practice for web-scale corpora, was a project-lead decision recorded in the source registry. Several sources were rejected, for example DepEd LRMDS (non-commercial), TLUnified (no licence, and contains evaluation sources), and OpenSubtitles (copyrighted).

3. The pipeline

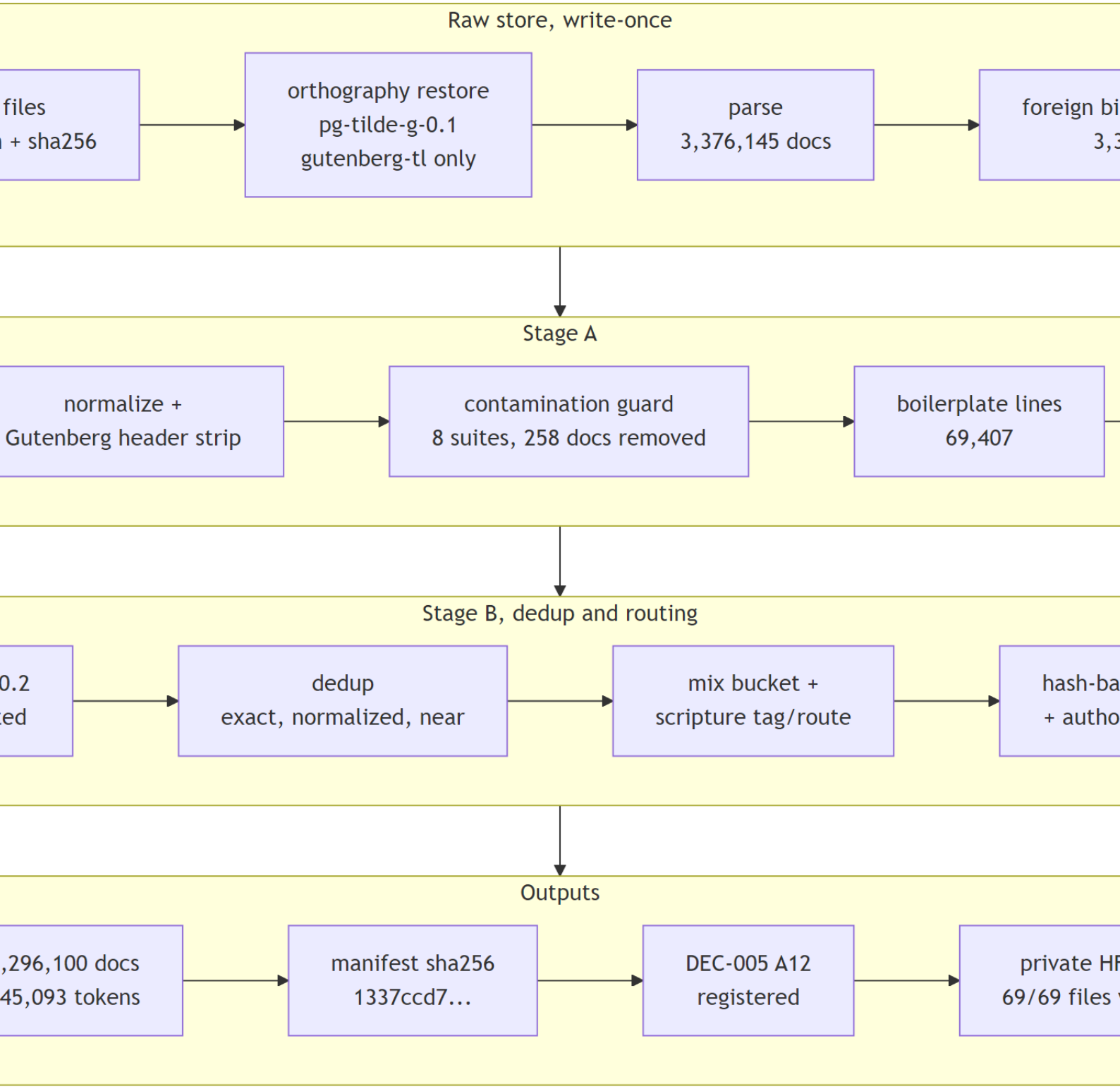


Figure 1: CORPUS-001: the PH-Corpus v0.1 r2 build, from pinned raw files to the registered, privately stored corpus. Counts are from the r2 manifest.

Table 2: Pipeline stages in the code's order, with their counters read from the r2 manifest. "A" and "B" are the pipeline's two cleaning stages.

stage	version	what it does	r2 counter
orthography	pg-tilde-g-0.1	restores the archaic g-tilde in the raw Gutenberg text (§3.1)	65,144 + 40 substitutions
parse	parsers-0.3	reads each source format into documents	parsed 3,376,145
foreign-Bible drop	bible-tagger-0.1	drops foreign-language biblica.com Bibles (FineWeb-2 regional only)	3,336 docs
A: domain blocklist	domains-0.1	drops `*.gov.ph` pages	45,516 docs
A: normalize	normalize-0.1	Unicode normalization; strips Gutenberg headers and licence boilerplate	
A: contamination	contamination-0.3	removes docs that share an n-gram (or, for short items, the whole item) with any of 57,315 eval items	258 docs
A: boilerplate	boilerplate-0.1	drops navigation and boilerplate lines	69,407 lines
repeated lines		removes lines repeated within a window	291 lines
B: PII redaction	pii-rules-0.2	replaces phone numbers, emails, government IDs, card numbers (Luhn-checked), IPs and secrets with typed placeholders	54,480 redactions
B: quality	quality-rules-0.2	Gopher-style rules (length, word length, symbols, bullets, ellipses, repeated lines, function words, benchmark formatting)	11,593 docs rejected
dedup	dedup-0.3	exact, normalised and near-duplicate removal	178 / 21 / 1,810
routing	bible-tagger-0.1	assigns the mix bucket; tags scripture and routes it to `regional_scripture`	14,568 tagged
split		hash-based holdout and author-level OOD (§3.3)	17,265 held out; 13 OOD
export	ph-corpus-pipeline-0.11	seeded global shuffle (seed 20261001), shard, write the manifest	3,296,100 train docs

PII redaction by type: phone 29,883; IP 9,579; government ID 9,532; email 3,807; card 1,363; secret 316. Quality rejections by reason: too short 4,911; ellipsis lines 3,074; bullets 1,249; benchmark format 1,219; non-alphabetic 377; no function words 297; symbols 253; word length 136; repeated lines 77.

3.1 Restoring the archaic g-tilde (pg-tilde-g-0.1)

Pre-war Tagalog spelled the *ng* sound with a tilde over the g (*ngg̃*, *mg̃a*). Project Gutenberg's plain-text files flatten it to `~g`. Rule pg-tilde-g-0.1 restores `~g` to g + U+0303 and `[)g]` to g + U+0306 (NFC turns the latter into ğ). It runs on the raw text before any other stage, so the training split and both evaluation splits see the same text.

- **Control.** A with/without test on the same raw store (`tests/data/test_orthography.py`). Without the rule, `~g` survives in train and in eval_ood. With it, none is left and the per-document counts are exact. The real-data control on all 32 Gutenberg texts gave 65,144 tilde_g + 40 breve_g substitutions, the same 32 documents kept, and 0 quality rejects either way. The r2 manifest records exactly these counts.
- **Evidence that the rule restores what was printed.** Two Internet Archive page scans of the same works show g̃ on the checked lines: *Dating Pilipinas* (1907) p.10, and a Tissot medical text p.34 (worn print; supporting evidence only).

- **Effect on tokens.** Gemma's tokenizer splits `n~g` into three tokens and `nǧ` into two, so the rule *reduces* Gutenberg tokens: 2,478,992 to 2,443,924 (-35,068).
- **Disclosed side effect.** Python's `\w` does not match U+0303, so the word statistics used by the quality rules split *mg̃a* into two words. No keep/reject decision changed on this data.
- **A measurement error, corrected.** The first token estimate (+1.35%) was wrong. A shell-escaping bug in the measurement script turned the regex replacement `\1` into a control byte. The corrected measurement used the real module. This is a post-mortem item (PM-017).

3.2 Scripture and language-ID findings

FineWeb-2's regional files contain a large share of Bible text, which would over-weight one register. The Bible tagger marks scripture by page address first (Bible-reader hosts and paths), with an inline verse-run fallback (at least 8 consecutive verse numbers). We found that chapter:verse density, the signal first proposed, misses reader pages: they print verse numbers inline and have about zero "3:16"-style references.

- **Precision**: 49/50 tagged documents were scripture (Wilson 95% 0.895-0.996). The one false positive was a magazine study article with numbered paragraphs. In the same check, 46/50 were strictly Bible (Wilson 95% 0.812-0.968), and 3 were LDS scriptures (Book of Mormon in Bikol ×2, Doctrine and Covenants in Cebuano). The URL rule tagged 46 documents and the verse-run fallback 4; all 4 fallback tags were the non-Bible ones (the 3 LDS texts and the false positive). That is why the bucket is named `regional_scripture`, not "Bible".
- **Recall**: 0/50 scripture documents were missed among untagged hil and pag documents (upper bound 0.071).
- **Measured share**: scripture is 10.54% of regional tokens, under the 15% trigger at which a separate cap would apply. Per language, the share is uneven: pag 45.8%, hil 38.7%, war 18.9%, bcl 18.3%, pam 15.9%, ilo 10.7%, ceb 5.3%.

Two language-ID findings:

- **Foreign Bibles.** In the tagged hand-check, 6/50 documents (0.056-0.238) were biblica.com Bibles in non-Philippine languages that FineWeb-2 labelled Hiligaynon. biblica.com regional documents are now kept only for an allow-list of Philippine version codes (`apsd-ceb`, `hlgn`); 3,336 documents across 61 foreign version codes were dropped.
- **Noisy labels.** A spot-check of 50 untagged hil and pag documents found 6/50 (0.056-0.238) in the wrong language. Only 2/50 (0.011-0.135) were non-Philippine; the other 4 were *other* Philippine languages under the wrong label (Aklanon, Kalanguya, Sambal, Ibaloi). No general language-ID filter was applied in v0.1, because the candidate tool was not yet cleared. **Per-language numbers for the small regional languages, including per-language evaluation slices, should be read as label-noisy.**

3.3 Holdout and out-of-distribution splits

The held-out split is assigned by hash: a document is held out when sha256(doc_id) mod 1000 < K for its language, with a floor of 200 documents and K capped at 50. The out-of-distribution (OOD) split holds every work by 5 Gutenberg authors chosen by the sha256 of the author name (13 books). These books never enter training.

Table 3: Held-out documents per language. pag, war and pam stay under the 200 floor at the K cap; their evaluation slices are small.

language	K	held-out docs
tl	5	11,582
en	5	3,747
ceb	5	1,016
ilo	11	225
hil	5	216
bcl	27	202
pag	50	118
war	50	81
pam	50	78
total		**17,265**

Because doc_id is a hash of the text, the g-tilde restoration changed the id of one Gutenberg book (ebook 14794). Its new hash fell under tl's K, so it moved from train (as-is build) to the tl holdout (r2). The arithmetic closes exactly on the real-data control:

- train, the 18 books that stayed: −11,271 tokens;
- eval_ood: −23,385 tokens;
- the moved book: −412 tokens;
- total: −35,068 tokens, the control's delta.

So the holdout now contains one archaic-orthography literary book. DEC-005 Amendment 12 pre-declared that the C1 primary held-out slice excludes held-out documents whose mix bucket is literary; the inclusive number is reported as a sensitivity row. Amendment 13 registered the scorer that applies this rule. Archaic literary text is measured on eval_ood, as registered.

4. Contamination guard

The guard removes any document that shares an n-gram (n = 13 for the external suites, 8 for PH-EVAL), or for short items the whole item, with an item from 8 evaluation suites, 57,315 items in total. An item with fewer than n words but at least the suite's minimum phrase length is matched as one whole-text phrase. It is a hash-only pickle (sha256 `eed9aaae…`), so no benchmark text ships with the build. The guard was applied at build time and removed 258 documents. The largest families were English MMLU 96, Global-MMLU English 96 (largely the same documents), FilBench INCLUDE 58 and HellaSwag 44.

Table 4: The 8 guard suites, from the r2 manifest. "Unprobeable" items are shorter than the suite's minimum phrase length (8 words; 5 for stingray), so the guard cannot test them.

suite	items	n	unprobeable
english	25,256	13	4
gmmlu_en	14,042	13	4
filbench	12,956	13	343
ntrex	3,994	13	276
coding	542	13	0
stingray	200	13	9
ph-eval	175	8	0
kalahi	150	13	0

An independent scan-only pass then re-checked every document in train, eval_holdout and eval_ood (3,313,378 documents) with the same guard. It found **0 hits in every family**. Known gap: the 9 unprobeable stingray items (1 Tagalog), which no n-gram guard can check.

5. Size and the registered training mixture

Table 5: Training buckets in r2 (Gemma tokenizer @411aa17b, tokenizer.json sha256 `12bac982…`).

bucket	docs	tokens	share of train tokens
tl	2,300,906	2,422,503,623	75.93%
en_replay	728,987	506,064,049	15.86%
regional	251,725	231,906,908	7.27%
regional_scripture	14,398	27,333,904	0.86%
literary	84	2,836,609	0.09%
total	**3,296,100**	**3,190,645,093**	

The literary bucket holds 84 literary documents (18 Gutenberg works + 66 books of Ang Biblia 1905). The other 14 of the 32 Gutenberg works are in evaluation: 13 in eval_ood and 1 (ebook 14794) in the tl holdout.

The corpus is larger than the pilot reads, so the mixture sets how often each bucket is sampled. The weights were computed from the measured counts and registered in Amendment 12:

Table 6: The registered pilot mixture (DEC-005 Amendment 12).

bucket	weight	projected epochs at 2,000,158,720 pilot tokens
tl	0.6445	0.532
en_replay	0.20	0.790
regional	0.134184	1.157
regional_scripture	0.015816	1.157
literary	0.0055	3.878

The weights were set as follows:

- **Regional** is split at its natural proportion: scripture is 10.54% of regional tokens, under the 15% trigger, so the mixture is the same as leaving scripture inside regional.
- **Literary** is capped by the 4-epoch rule: floor4(3.9 × 2,836,609 / 2,000,158,720) = 0.0055. The factor 3.9 rather than 4 leaves a margin for skipped batches and the synthetic BOS token.
- **Control.** The real trainer gate refused the pilot config as first written (literary 0.03, which would have meant 21.152 epochs) and passed the registered mixture.

6. Storage, registration and cost

- **Manifest**: sha256 `1337ccd7332c91a243d01d968b6cac78b4d117ee9dc3f03cec1bcd4c654e5e6b`, hashed twice (after download and on the local copy). 66 train parts + 2 evaluation parts.
- **Registration**: DEC-005 Amendment 12 (2026-10-05T09:35Z). The earlier as-is build (no g-tilde restoration) is measurement-only and never registered or uploaded.
- **Private upload**: `gryphlee/ph-corpus` (private asserted before writing) at commit `d6f1bfea5b3a9b719d84b698f7cfda72a9a79d99`, folder `PH-Corpus-v0.1/`. All 69 files (66 train parts, 2 evaluation parts, `manifest.json`) were verified by sha256 at that exact commit. A streaming reload with `datasets` returned 3,296,100 rows against 3,296,100 expected.
- **Reliability fixes made along the way**:
  - two earlier attempts died to cloud preemption, so builds now publish atomically and run nonpreemptible;
  - one launch was cancelled when the laptop lost DNS mid-run, so long builds now run as a deployed app with a server-side spawn, which a dropped laptop connection cannot cancel. These are written up in PM-017.
- **Compute cost**: measured from wall time, as-is $1.412 + r2 $4.591 + scan $0.368 = $6.371. Estimated aborted attempts bring this to about $8.7. The provider dashboard is authoritative.

7. Privacy and data-protection status

The web sources contain incidental personal data (names in public pages; contact details before redaction). Following NPC Advisory No. 2026-01, governance drafted a lawful-basis memo (legitimate interest, RA 10173 s.12(f)) and a pre-filled privacy impact assessment. **Both are drafts, pending review by the Kooya Data Protection Officer (DPO).** The declared purpose, quoted verbatim from the memo (s.2.1), is:

"PH-Corpus is used only to train and evaluate Philippine-language AI models."

Table 7: Privacy paperwork. Each hash is printed in two halves of 32 hex characters for width; join them without a space.

document	sha256	status
`data/registry/compliance/lawful-basis-memo-ph-corpus-web.md`	`96c924c0a92a520f9201b30720c0fb81` `482511b9dbe03c79ec113006dadae5e0`	DRAFT, pending DPO review
`data/registry/compliance/pia-template-ph-corpus-web.md`	`765494592316488a43fcd49a9c6f6aa1` `a7000324b352a6d28c4a479918ea6822`	TEMPLATE, pending DPO review

The memo's own caveat applies: the advisory's primary text had not yet been read, because automated fetch was blocked. The memo must be re-checked against it. Open items for the DPO include the controller's legal entity name, a contact point for data-subject requests, and the cross-border transfer basis for the private storage.

Scope note: r2 excludes `*.gov.ph` text (45,516 documents blocked). The project lead's 2026-10-05 decision to admit `*.gov.ph` public text with attribution applies only to the future v1.0 PH-knowledge stage, not to this corpus.

8. Limitations

1. **Not public.** The corpus is private; this paper is internal. Any public release needs its own decision and the DPO review above.
2. **Label-noisy regional languages.** 6/50 spot-checked hil/pag documents carried the wrong language label (§3.2). Per-language results for small regional languages are indicative only.
3. **Small regional evaluation slices.** pag, war and pam have 78-118 held-out documents (Table 3).
4. **Web text dominates.** 75.93% of training tokens are Tagalog web text from one crawl family. Register and topic balance were not measured beyond the quality filters; a blind keep/reject audit by native raters is pending.
5. **Guard blind spot.** The guard cannot test items shorter than the minimum phrase length (8 words; 5 for stingray): 9 stingray, 343 FilBench and 276 NTREX items are unprobeable. It does not catch paraphrases.
6. **Owner rights in crawled pages.** The packaging licences do not transfer the rights in each page's text (§2).

9. Reproducibility

Table 8: Files.

item	path
manifest	`data/processed/ph_corpus/PH-Corpus-v0.1-r2/manifest.json`
build registry	`data/processed/ph_corpus/BUILDS.yaml`
source registry and licences	`data/registry/corpus_sources.yaml`
build, stage and upload on Modal	`scripts/data/modal_corpus.py`, `scripts/data/upload_corpus.py`
orthography rule and test	`src/phfm/data/cleaning/orthography.py`, `tests/data/test_orthography.py`
Bible tagger	`src/phfm/data/quality/bible.py`
PII rules	`src/phfm/data/privacy/pii.py`
registration	`docs/DECISION_LOG.md`, DEC-005 Amendments 12 and 13

Table 9: Hashes, printed in two halves of 32 hex characters; join them without a space.

artifact	sha256
manifest.json	`1337ccd7332c91a243d01d968b6cac78` `b4d117ee9dc3f03cec1bcd4c654e5e6b`
manifest config	`8b241ce274c68afeb378fca5667a731f` `577730d566c61a8c1f98ed5b0f0fcfaa`
contamination guard pickle	`eed9aaaeabae74bcd1442e57fd00770d` `7fbcbf544c91ea1371d796db496e2dde`
eval_holdout part	`d0a35f62d1bae80b28e9866e8df86f75` `01ae8eaafd421e41c7cb8d4c32047a1b`
eval_ood part	`e631e917ff97ef2709c58096c71f8d87` `7f786527492256d94a97c95526d7d64c`
tokenizer.json (gemma-4-E4B @411aa17b)	`12bac982b793c44b03d52a250a9f0d0b` `666813da566b910c24a6da0695fd11e6`

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